RESEARCH PROPOSAL • MASTER'S DISSERTATION FOUNDATION

In-Silico Multimodal Closed-Loop Emotional Control in 2D Valence–Arousal Space: Fusing Sparse Wearable EEG and Autonomic Signals with Multi-Actuator Model Predictive Control

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Target Venue: *Biomedical Signal Processing and Control* (Elsevier, Q1/Q2, IF ~4.9)

Target Completion: Late December 2026 (8 Weeks)

Publishing Track: Standard Subscription ($0 APC) / Institutional OA

1. Executive Summary & Strategic Motivation

This study establishes the mathematical, state-estimation, and control-theoretic core of the Master's Dissertation on **Closed-Loop Emotional Control Using Multimodal Sensors**. Rather than risking a rushed human recruitment campaign within an 8-week window (November–December), this paper adopts the validated ***in silico* (computer-simulated digital twin) paradigm** established in biomedical control engineering (e.g., Fekri Azgomi & Faghih, 2022, *IEEE OJEMB*).

> **Core Idea:** We identify subject-specific 2D state-space "Virtual Subjects" directly from two landmark public multimodal datasets—**DEAP** (N = 32; Koelstra et al., 2012) and **Emognition** (N = 43; Saganowski et al., 2022)—using a sparse, wearable-compatible sensor subset (OpenBCI EEG channels + wrist EDA + PPG/HRV). We then couple these virtual subjects with empirically validated models of three non-invasive actuators (**Generative Music, Ambient Lighting, and Wrist Heartbeat Haptics**) to benchmark **Model Predictive Control (MPC), Fuzzy-Supervised MPC, and Reinforcement Learning (RL)** across all four quadrants of Russell's Circumplex Model of Affect.

Three Novel Contributions Over Existing Literature

1. **From 1D Stress to 2D Valence–Arousal State-Space Control:** Prior state-space closed-loop simulations (Fekri Azgomi & Faghih, 2022; Khazaei et al., 2024) regulated only 1D arousal from a single sensor (EDA). We formulate and control the full 2D continuous state vector $\boldsymbol{x_k} = [\text{Valence}_k, \text{Arousal}_k]^T$.
2. **Hysteresis-Aware Multimodal Sensor Fusion (Sparse EEG + EDA + HRV):** Cortical EEG responds rapidly to valence and cognitive effort, whereas peripheral autonomic signals (EDA, HRV) exhibit slow sympathetic carry-over ("affective priming" / excitation transfer; Gutierrez Maestro et al., 2025; Fernández-Aguilar et al., 2019). Our Kalman/Bayesian observer explicitly accounts for emotional inertia and compensates for transition hysteresis.
3. **Realistic Coupled Multi-Actuator Allocation:** Instead of assuming an idealized scalar control input u_k, we model real actuator constraints and cross-dimensional coupling—specifically how **Lighting** (Li et al., 2025) and **Haptics** (Azevedo et al., 2017) compensate for the **asymmetrical tempo–valence crossover** in real-time generative music (Agres et al., 2023, 2025), where slow calming music inadvertently dampens positive valence.

2. Closed-Loop System Architecture

1. Virtual Subject (Plant)
Identified from **DEAP** (N=32) & **Emognition** (N=43).
Hidden 2D state $\boldsymbol{x_k} = [\boldsymbol{V_k}, \boldsymbol{A_k}]^T$ with emotional inertia $\boldsymbol{A_i}$ and disturbances $\boldsymbol{d_k}$.

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2. Multimodal Sensors
- **Sparse EEG:** FAA (F4–F3 α), Fz θ, P4 α, T7/T8 β/γ, phase coherence.
- **Peripheral ANS:** Deconvolved EDA (SCR) + PPG/HRV (RMSSD, HF, LF/HF).

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3. State Estimator
Affective Priming Score (APS) filter + **Multimodal Kalman / Bayesian Observer** outputting $\hat{\boldsymbol{x}}_k = [\hat{V}_k, \hat{A}_k]^T$ every 2–4 s.

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4. Controller & Actuators
Fuzzy-MPC / RL computes optimal $\boldsymbol{u_k}$:
1. **Music** (Tempo, Mode, Pitch)
2. **Light** (Color × Lux)
3. **Haptics** (–20% HR pulse)

3. Mathematical Formulation Summary

3.1 Virtual Subject State-Space Model & Multimodal Observation

For each participant *i*, discrete-time affective dynamics (Δt = 2–4 s) and multimodal observations are modeled as:

$$x_{k+1} = A_i x_k + B_i g(u_k) + d_k + w_k, \qquad y_k = C_i x_k + v_k$$

- **State Vector $x_k \in [0, 1]^2$:** Continuous normalized Valence and Arousal, with subject-specific inertia matrix $A_i \in \Re^{2\times2}$ capturing sequential carry-over.
- **Observation Vector $y_k \in \Re^m$:** Concatenates OpenBCI-compatible EEG features (Frontal Alpha Asymmetry, Fz theta, P4 upper alpha, Mean Phase Coherence; Ramirez et al., 2015; Ewing et al., 2016; Marín-Morales et al., 2018) with autonomic features (phasic SCR impulses and HRV; Koelstra et al., 2012; Khazaei et al., 2024).

3.2 Empirically Grounded Tri-Modal Actuator Model $g(u_k)$

Actuator Modality	Control Parameters u_k	Empirical Response & Constraints from Literature
1. Generative Music (Ehrlich 2019; Agres 2023, 2025)	Target musical valence & arousal: $u_{mus,V}$, $u_{mus,A} \in [0, 1]^2$ (Mode, Pitch, Tempo, Velocity)	Strong linear fit (R^2 = 0.90–0.96), but exhibits an **asymmetrical tempo–valence crossover**: low arousal settings ($u_{mus,A} < 0.4$, slow tempo) dampen perceived positive valence.
2. Ambient Lighting (Li, Wang, & Whang, 2025)	6 Colors (White, Purple, Blue, Green, Yellow, Red) × 3 Illuminances (20, 100, 180 lx)	**Green 180 lx** maximizes positive valence (*M*=3.66) & cognitive reappraisal (FMT/TBR); **Purple 20 lx** induces calm positive affect (FAA β=+0.049, FMT β=+0.589); **Red 180 lx / Blue** boost arousal (β=+0.792).
3. Wrist Haptics (Azevedo et al., 2017)	Vibrotactile heartbeat frequency: $u_{hap} \in [0, 1]$ (−20% resting HR, clamped to 40–65 bpm)	**Unidirectional calming actuator ($\Delta A < 0$):** Significantly reduces sympathetic EDA arousal (Hedges' g = 0.54, p = 0.029) and state anxiety (g = 0.77, p = 0.007) without cognitive load.

3.3 Closed-Loop Controllers Evaluated

We benchmark four controllers across the four clinical/therapeutic trajectories defined by Daly et al. (2016)—**De-stress** (High *A*, Low *V* → Neutral), **Calm** (High *A* → Low *A*), **Make Happy** (Low *V* → High *V*), and **Excite** (Low *A* → High *A*):

1. **Open-Loop Static Stimulation** (Baseline reference).
2. **Linear Quadratic Regulator (LQR)** with fixed state/effort weights (Q, R).
3. **Constrained MPC & Fuzzy-Supervised MPC:** Optimizes multi-step trajectory over horizon N_p subject to actuator bounds and music crossover coupling, with a Mamdani fuzzy supervisor dynamically adapting (Q_k, R_k) from tracking error $||e_k||$ (Fekri Azgomi & Faghih, 2022).
4. **Reinforcement Learning (DQN / Soft Actor-Critic):** Learns an adaptive multi-actuator policy resilient to across-session parameter drift (Humaidan et al., 2026).

4. Manuscript Blueprint for *Biomedical Signal Processing and Control*

Target Scope: Full-Length Research Article (~6,000 words, 9–10 pages in 2-column format, 6 Figures, 3 Tables, ~40 References).

- **Section 1 — Introduction (~900 words):** Motivation for closed-loop affective CPS; limitations of 1D single-sensor loops; contributions (**Fig. 1:** Closed-loop block diagram).
- **Section 2 — Materials & Methods (~2,200 words):** DEAP & Emognition preprocessing; OpenBCI sparse channel mapping; Affective Priming Score (APS) filtering; 2D state-space identification & Kalman filter; coupled tri-modal actuator equations; LQR, Fuzzy-MPC, and RL formulations.
- **Section 3 — *In Silico* Experiments & Results (~1,600 words):**

- *Exp. 1 (State Estimation):* Sparse EEG vs. ANS vs. Multimodal Fusion accuracy & priming error reduction (**Table 1, Fig. 2**).
- *Exp. 2 (4-Quadrant Closed-Loop Control):* Settling time, RMSE, and control effort across *De-stress, Calm, Make Happy, Excite* (**Table 2, Figs. 3–4**).
- *Exp. 3 (Actuator Synergy & Sensor Dropout Robustness):* How Lighting + Haptics overcome the Music tempo–valence crossover, and Kalman resilience to simulated EEG motion artifacts (**Table 3, Figs. 5–6**).

- **Section 4 & 5 — Discussion & Conclusion (~1,150 words):** Direct hardware translation path to the OpenBCI prototype for the Master's Dissertation; neuroethical user "veto control" (Steinert & Friedrich, 2020).

5. 8-Week Execution Roadmap (Nov 1 – Dec 27, 2026)

Week	Dates	Key Milestone & Concrete Deliverable
Week 1	Nov 1 – Nov 8	**Data Pipeline & Feature Extraction:** Extract sparse EEG (Fz, F3, F4, P4, T7, T8) + EDA + HRV features on DEAP/Emognition; implement APS transition filter.
Week 2	Nov 9 – Nov 15	**System ID & Multimodal Kalman Observer:** Fit 2D virtual subject models; benchmark EEG-only vs. ANS-only vs. Multimodal fusion (**Table 1 & Fig. 2**).
Week 3	Nov 16 – Nov 22	**Tri-Modal Actuator Simulation:** Code coupled Music (Agres 2023/2025), Lighting (Li 2025), and Haptics (Azevedo 2017) response surfaces in Python.
Week 4	Nov 23 – Nov 29	**Controller Implementation:** Implement LQR, Constrained MPC, Fuzzy-MPC, and RL controllers across the 4 therapeutic quadrant transitions.
Week 5	Nov 30 – Dec 6	**Closed-Loop & Robustness Benchmarks:** Run Monte Carlo simulations & sensor-dropout stress tests; generate final publication plots (**Tables 2–3, Figs. 3–6**).
Week 6	Dec 7 – Dec 13	**Manuscript Drafting (Part 1):** Draft Section 2 (Methods) and Section 3 (Results) in the official Elsevier LaTeX template.
Week 7	Dec 14 – Dec 20	**Manuscript Drafting (Part 2):** Draft Introduction, Discussion, Conclusion, Abstract (200 words), and BSPC Highlights (5 bullets ≤ 85 chars).
Week 8	Dec 21 – Dec 27	**Supervisor Review & Submission Prep:** Integrate supervisor feedback, polish figures & GitHub code repository, and finalize BSPC submission package.